Is Semantic Level of Detail Linearly Decodable from Frozen SciBERT Embeddings?

Part 3 — Evaluation & Technical Report

Introduction

Semantic Level of Detail (SLoD) distinguishes high-level scientific statements (**macro**), section-level framing (**meso**), and implementation or result details (**micro**). For the Temporal Knowledge Hypergraph, SLoD could route retrieval toward summaries, section overviews, or fine-grained evidence. The research question is whether SLoD is linearly encoded in frozen scientific-text embeddings.

Methodology

Weak labels. Rule-based weak supervision mapped QASPER structure to labels: title/abstract/first two introduction paragraphs/conclusion → macro; first sentence of eligible section leads → meso; non-lead methods, experiments, results, evaluation, implementation, and approach text → micro. The balanced dataset has 5,292 spans (1,764/class) from 877 NLP papers. Paper-ID splits (70/10/20) prevent leakage; normal spans contain 30–300 tokens and use per-paper/per-label caps.

Representations and probe. Frozen *allenai/scibert_scivocab_uncased* token states were attention-mask-aware mean pooled; weights were never updated. Logistic regression was trained on cached vectors. Baselines predict the majority training class or use only section_name. The separate length-control dataset truncates every span to exactly 30 tokens before re-embedding and repeats the same paper-grouped evaluation.

Experimental conditions

Condition	Train/test	Purpose
In-domain	NLP → held-out NLP papers	Primary linear decoding test
Cross-domain	N/A	Second domain unavailable
Length-controlled	NLP → held-out NLP papers	Remove span-length variation

Quantitative results

Condition / model	Accuracy	Macro-F1	Status
In-domain embedding	0.886	0.886	Complete
Majority baseline	0.316	0.160	Complete
Section-name baseline	0.788	0.785	Complete
Length-controlled embedding	0.779	0.780	Complete
Cross-domain	N/A	N/A	Only QASPER/NLP available

Length control reduced macro-F1 by **0.105** (10.5 points), yet controlled performance remained far above majority. Length is therefore a substantial confound. The section-name baseline (0.785) slightly exceeds the controlled embedding probe (0.780), so the present evidence does not isolate embedding-specific semantic abstraction from structural/lexical section cues.

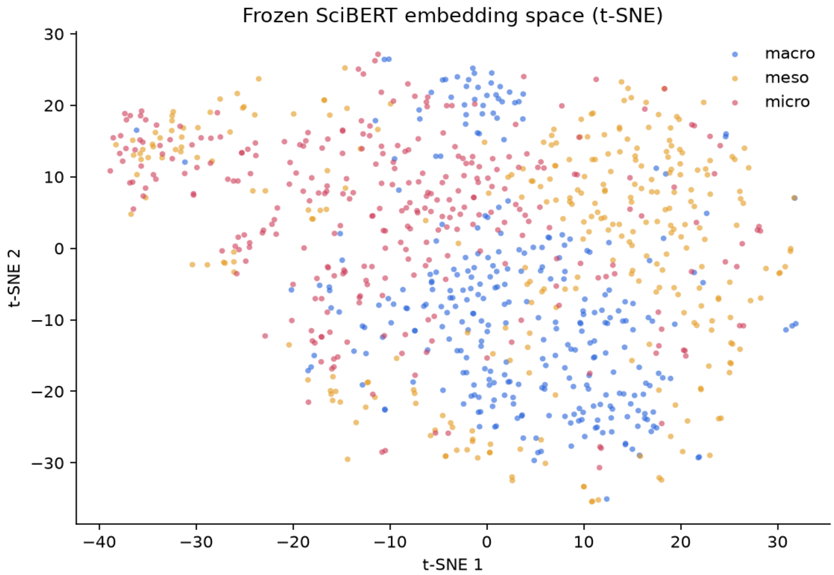
Confusion matrices

In-domain

	Pred macro	Pred meso	Pred micro		Pred macro	Pred meso	Pred micro
True macro	333	15	23	True macro	285	23	63
True meso	6	314	17	True meso	24	280	33
True micro	38	23	299	True micro	60	33	267

Left: in-domain. Right: exact-length controlled.

The main normal error was micro→macro (38). Under control, macro↔micro confusion dominated (63/60). Meso was not hardest: normal F1=0.912; micro was hardest (0.856 normal, 0.739 controlled). Cross-domain generalization cannot be inferred without a second domain.



t-SNE: partial class structure with substantial overlap; descriptive, not a linear-separability test.

High-confidence correct examples

Three examples per true class; confidence is the probe's maximum predicted probability.

MACRO

1. macro→macro (1.000) — In this work we proposed a new annotation scheme for three theoretically-motivated features of student talk in classroom discussion: argumentation, specificity, and knowledge domain. We dem…

2. macro→macro (1.000) — News website comment sections are spaces where potentially conflicting opinions and beliefs are voiced. Addressing questions of how to study such cultural and societal conflicts through tec…

3. macro→macro (1.000) — Closely modelling questions could be of importance for question answering and machine reading. In this paper, we introduce syntactic information to help encode questions in neural networks.…

MESO

1. meso→meso (1.000) — We use Praat's autocorrelation-based pitch estimation algorithm to extract the fundamental frequency (F0) contour for each sample, using a minimum frequency of 75Hz and a maximum frequency…

2. meso→meso (1.000) — In this section, we introduce the two data sets we used for our revision detection experiments: Wikipedia revision dumps and a document revision data set generated by a computer simulation.

3. meso→meso (1.000) — We begin by introducing the datasets (Section SECREF15), followed by detailing the featurization methods (Section SECREF17), the experiments (Section SECREF22), and finally reporting res…

MICRO

1. micro→micro (1.000) — where M is the average of the two distributions ($M=0.5*(A+B)$). Table TABREF10 shows the results for the four features we previously presented. Among others it can be noticed that the lo…

2. micro→micro (1.000) — where INLINEFORM0 denotes cross-entropy and INLINEFORM1 denotes KL-divergence. INLINEFORM2 corresponds to the learned parameters. INLINEFORM3 is a scalar that weights the loss terms. This s…

3. micro→micro (1.000) — Examining the results for English-or, we find that all models demonstrate humanlike expectation in the pl_or_pl and sg_or_pl conditions. The LSTM (1B), LSTM (PTB), and RNNG models show zero…

Interpretation

Correct macro examples use contribution/conclusion language; meso examples frame section contents or processing choices; micro examples contain equations, model parameters, or fine-grained result comparisons. These cues are compatible with SLoD, but also with document genre conventions.

High-confidence failures

MACRO

1. macro→meso (0.995) — Second, as mentioned earlier, we are in need of a fully (rather than partially) checked SICK dataset to examine the impact of data quality on the results since the parti…

2. macro→meso (0.992) — We gratefully acknowledge the many other contributors to this work, including Niranjan Balasubramanian, Matt Gardner, Peter Jansen, Jayant Krishnamurthy, Souvik Kundu, T…

3. macro→micro (0.991) — The word at the begin of the sentence is capitalized, so, if the name of person is at this position, model will ignore them (no extraction). To improve this issue, we ca…

MESO

1. meso→micro (0.998) — The following words were considered as stopwords in our analysis: all, just, don't, being, over, both, through, yourselves, its, before, o, don, hadn, herself, ll, had,…

2. meso→macro (0.956) — In this paper we have provided a unified account of referring expression choice that solves a long-recognized puzzle for rational theories of language use: why do speake…

3. meso→macro (0.937) — The construction of causal attribution networks generates important knowledge networks that may inform causal inference research and even help future AI systems to perfo…

MICRO

1. micro→meso (1.000) — Elisabot is composed of two models: the model in charge of asking questions about the image which we will refer to it as VQG model, and the Chatbot model which tries to…

2. micro→meso (0.997) — There are a few reasons that contributed to the low accuracy obtained on this task by various methods (including ours) compared to other NLP problems: weak supervision,…

3. micro→macro (0.996) — The University of Alberta (UAlberta) performed a focused investigation on four language pairs, training cognate-projection systems from external cognate lists. Two metho…

Discussion and future work

SLoD is **strongly linearly decodable in-domain under the current weak labels**, but this is not yet proof of a pure semantic hierarchy representation. Length and section-derived lexical structure explain substantial signal. In RAG, use the calibrated probe as an auxiliary router—macro for overviews, meso for section navigation, micro for implementation/numerical evidence—combined with metadata and uncertainty thresholds.

Next experiments: obtain a second domain; build a small human-labelled SLoD test set; mask section-title expressions; perform tighter content/length matching; compare layers and embedding models; add random-label/selectivity controls; and evaluate calibration and nonlinear probes. These would distinguish semantic abstraction from document-template artifacts.